

KYUSYUR, Russian Federation

WMO: 219210

Lat: 70.683N

Lon: 127.400E

Elev: 33

StdP: 100.93

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -49.9	(c) -48.5	(d) -51.5	(e) 0.0	(f) -49.4	(g) -50.2	(h) 0.0	(i) -48.3	(j) 13.1	(k) -24.5	(l) 11.5	(m) -25.3	(n) 0.7	(o) 50	(p) 0.919

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	24.8	16.3	22.3	15.4	19.9	14.2	17.6	22.7	16.2	21.2	14.7	19.2	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.1	19.8	13.8	9.9	18.8	12.2	8.9	17.4	50.0	23.1	45.8	21.3	41.5	19.3	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.4	(b) 9.7	(c) 8.5	(e) -51.5	(f) 29.1	(g) 1.9	(h) 2.1	(i) -52.8	(j) 30.6	(k) -53.9	(l) 31.8	(m) -55.0	(n) 33.0	(o) -56.3	(p) 34.5		
(a) 11.4	(b) 9.7	(c) 8.5	(e) -51.7	(f) 19.6	(g) 1.9	(h) 1.2	(i) -53.1	(j) 20.5	(k) -54.2	(l) 21.2	(m) -55.3	(n) 21.9	(o) -56.7	(p) 22.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.2	-36.1	-34.8	-25.1	-12.8	-2.1	9.9	12.5	10.5	3.2	-10.4	-27.7	-35.4
	DBStd	19.41	7.73	8.18	8.88	6.75	5.79	5.55	4.68	3.64	3.80	6.98	8.38	8.13
	HDD10.0	8335	1428	1255	1089	683	374	69	25	38	206	631	1130	1407
	HDD18.3	11172	1686	1488	1347	933	632	261	192	244	454	889	1380	1666
	CDD10.0	223	0	0	0	0	1	65	101	53	2	0	0	0
	CDD18.3	17	0	0	0	0	0	7	10	1	0	0	0	0
	CDH23.3	138	0	0	0	0	0	54	81	3	0	0	0	0
	CDH26.7	24	0	0	0	0	0	8	16	0	0	0	0	0
Wind	WSAvg	3.1	2.1	2.3	2.8	3.7	4.2	4.1	3.9	3.5	3.3	3.2	2.1	1.9
Precipitation	PrecAvg	343	15	16	17	15	19	36	52	54	41	41	22	18
	PrecMax	458	42	41	55	58	51	74	161	146	91	78	54	40
	PrecMin	215	1	1	2	0	1	7	10	13	8	13	2	5
	PrecStd	66	9	10	12	13	13	17	30	29	24	18	13	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-14.4	-11.2	-1.3	5.8	16.5	27.7	28.6	23.5	16.8	2.0	-4.6	-14.4
		MCWB	-14.9	-12.0	-2.6	2.2	9.2	16.9	18.4	16.3	11.0	0.7	-5.3	-14.9
	2%	DB	-19.1	-17.0	-4.7	2.7	11.4	24.8	25.6	21.2	12.7	0.3	-9.2	-17.5
		MCWB	-19.5	-17.5	-5.6	-0.1	6.3	15.4	17.5	15.0	8.5	-0.7	-9.8	-17.9
	5%	DB	-21.8	-20.0	-7.8	0.4	8.4	22.1	22.9	19.1	10.4	-0.7	-12.6	-20.2
		MCWB	-22.1	-20.5	-8.8	-1.8	4.5	14.3	16.6	14.1	7.4	-1.8	-13.0	-20.6
	10%	DB	-24.4	-22.5	-11.2	-2.1	6.0	19.2	20.3	16.8	8.7	-1.9	-15.7	-23.1
		MCWB	-24.7	-22.8	-12.0	-4.0	2.9	12.9	15.0	13.1	6.2	-2.9	-16.1	-23.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-14.9	-11.9	-2.5	2.2	9.5	17.9	19.7	17.9	11.8	0.8	-5.2	-14.8
		MCDB	-14.5	-11.2	-1.4	5.4	16.1	24.7	26.1	21.1	15.3	1.7	-4.7	-14.6
	2%	WB	-19.5	-17.4	-5.4	0.2	6.5	16.3	18.3	16.1	9.4	-0.4	-9.7	-18.0
		MCDB	-19.1	-17.0	-4.7	2.4	10.7	23.2	23.7	19.7	11.8	0.3	-9.2	-17.6
	5%	WB	-22.1	-20.4	-8.8	-1.7	4.7	14.9	17.2	14.7	7.8	-1.7	-13.0	-20.6
		MCDB	-21.8	-20.1	-7.8	0.3	8.2	20.8	22.0	18.1	10.0	-0.7	-12.6	-20.2
	10%	WB	-24.7	-22.8	-12.0	-4.0	3.1	13.1	15.6	13.4	6.4	-2.9	-16.0	-23.3
		MCDB	-24.5	-22.5	-11.3	-2.2	5.9	19.0	20.1	16.4	8.4	-1.9	-15.7	-23.1
Mean Daily Temperature Range	5% DB	MDBR	7.0	6.7	8.7	9.9	6.9	8.2	8.1	7.5	5.7	5.5	7.0	6.6
		MCDBR	9.3	9.1	8.6	11.0	9.7	12.2	11.7	11.6	9.1	4.4	8.9	9.9
		MCWBR	9.0	8.8	7.8	8.9	5.9	5.8	6.1	6.7	5.8	3.9	8.6	9.4
	5% WB	MCDBR	9.0	8.7	8.5	10.8	8.9	11.2	10.5	9.8	8.1	4.5	8.9	9.5
		MCWBR	8.8	8.5	7.8	8.9	5.6	6.2	6.2	6.4	5.4	4.2	8.6	9.3
Clear-Sky Solar Irradiance	taub	N/A	0.210	0.285	0.316	0.349	0.351	0.415	0.395	0.299	0.219	N/A	N/A	
	taud	N/A	1.988	2.033	2.066	2.153	2.411	2.236	2.245	2.419	2.033	N/A	N/A	
	Ebn at Noon	0	485	683	795	817	838	750	711	709	446	0	0	
	Edh at Noon	0	38	82	112	118	95	110	96	58	34	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.32	1.87	4.33	5.85	5.61	4.95	3.28	1.59	0.45	0.00	0.00	
	RadStd	0.00	0.04	0.13	0.20	0.36	0.54	0.35	0.42	0.21	0.07	0.00	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+1.05	N/A	N/A	N/A	N/A	N/A	-413	-394	N/A	N/A

Nomenclature: See separate page